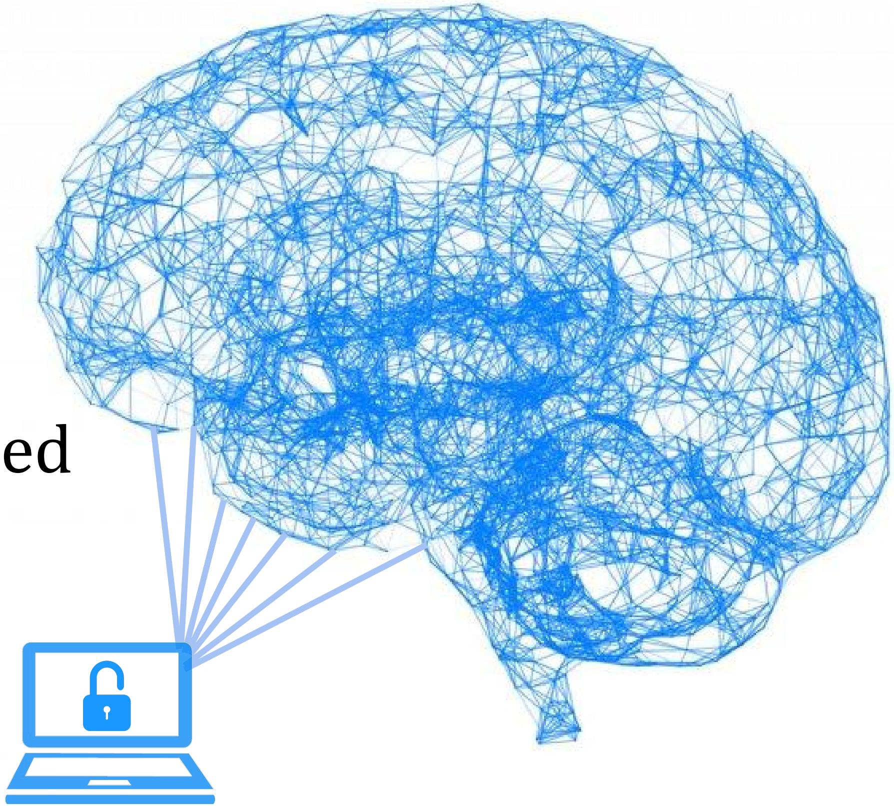
Brain ID:

Development of an EEG-Based Biometric Authentication System



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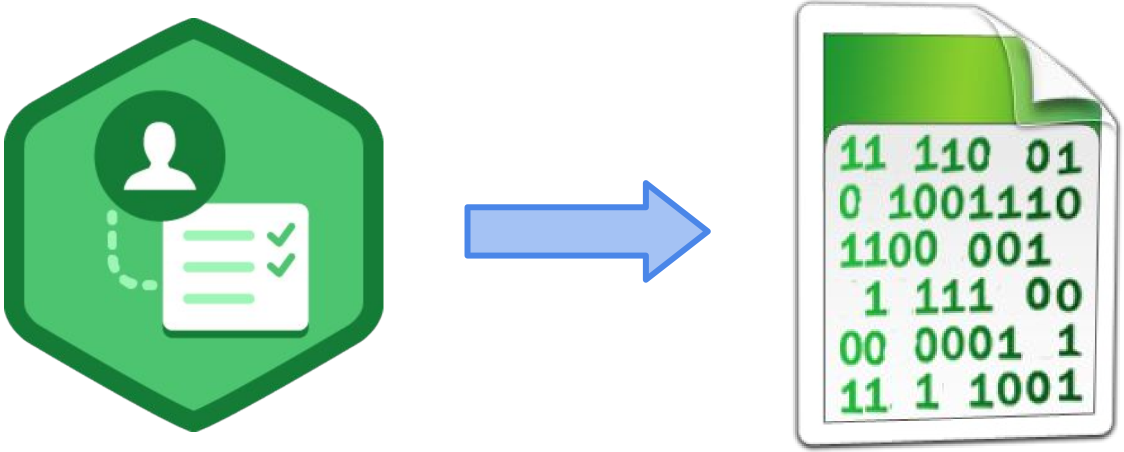
Content

Introduction

- Authentication: process to verify that a user has privilege to access secure information



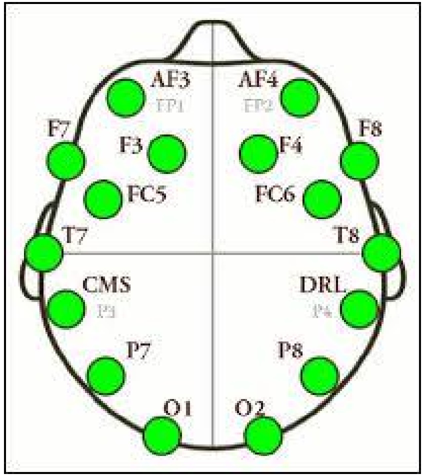
Introduction (Cont.)

- Authentication strategies
    - Knowledge (passwords, PIN, ...)
    - Possession (ATM card, passport, ...)
    - Biometric (Fingerprint, voice, iris, ...)

Introduction (Cont.)

- Electroencephalogram (EEG)
    - Electrophysiological monitoring method to record electrical activity of one’s brain
- EMOTIV Epoch+ headset

Headset



Top-down view of electrode placement

Past Research

- Person identification based on parametric processing of EEG by Poulos et al., 1999
- Low-cost EEG-based authentication by Ashby et al., 2011
- Exploring EEG-based biometrics for user identification and authentication by Gui et al., 2014

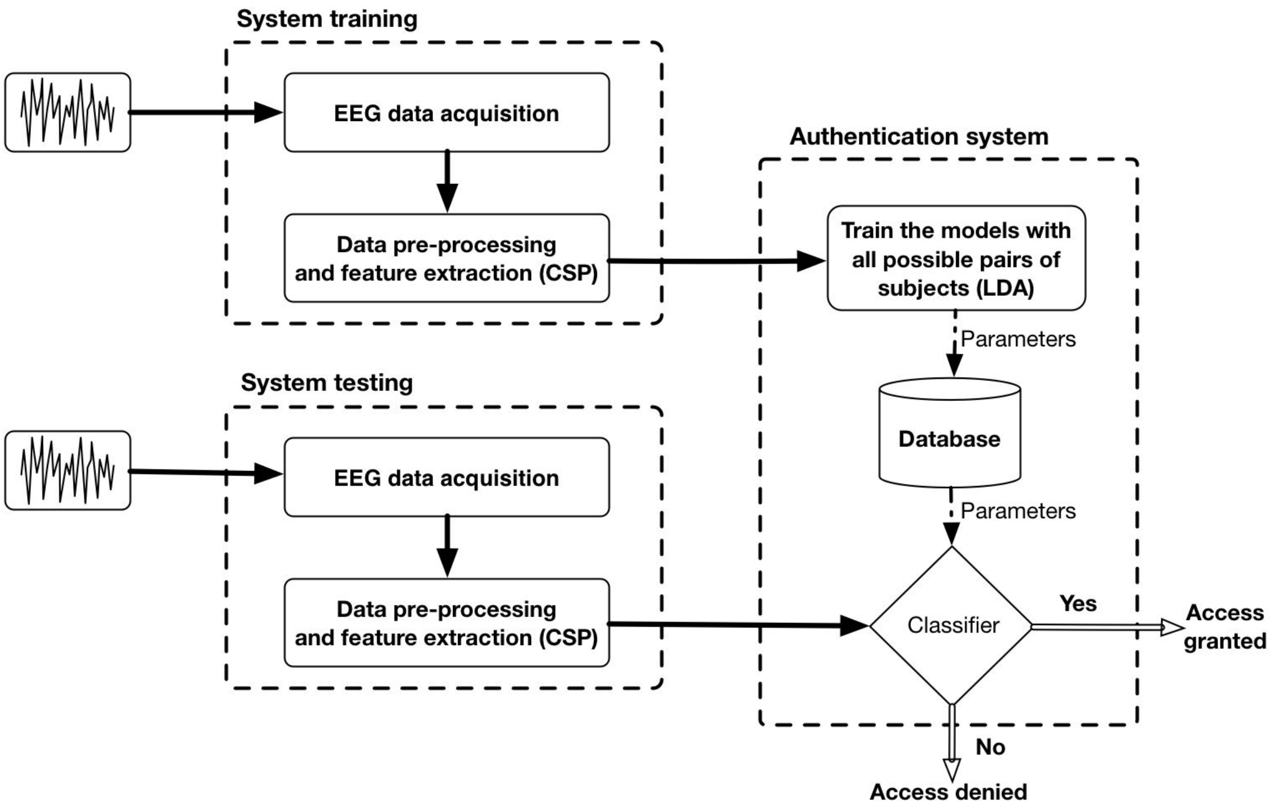
Motivation

- Privacy problems in knowledge-type strategies
- Traditional biometric systems can be misused by coercion
- Different EEG signal patterns emit for different mental states

Goal

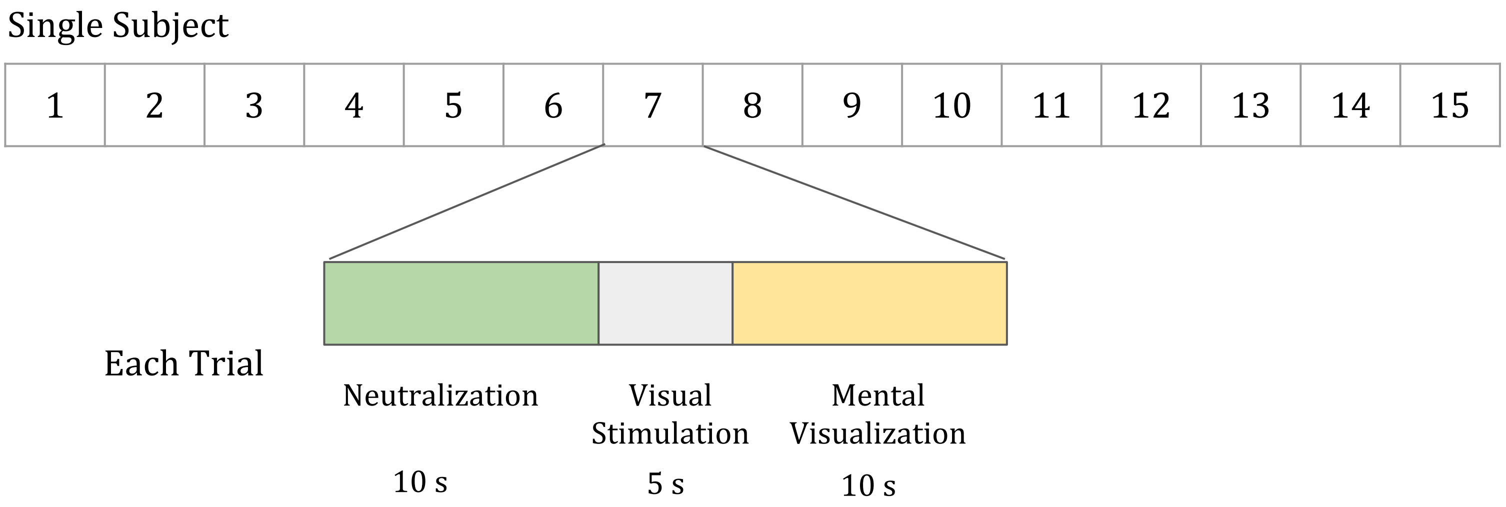
- Develop a robust authentication system to secure information systems using metal visualization confirmation

Methodology



Methodology (Cont.)

- Task
  - 12 subjects
  - 15 trials conducted for each subject
  - Random 4-digit numbers were used for visual stimulation



Methodology (Cont.)

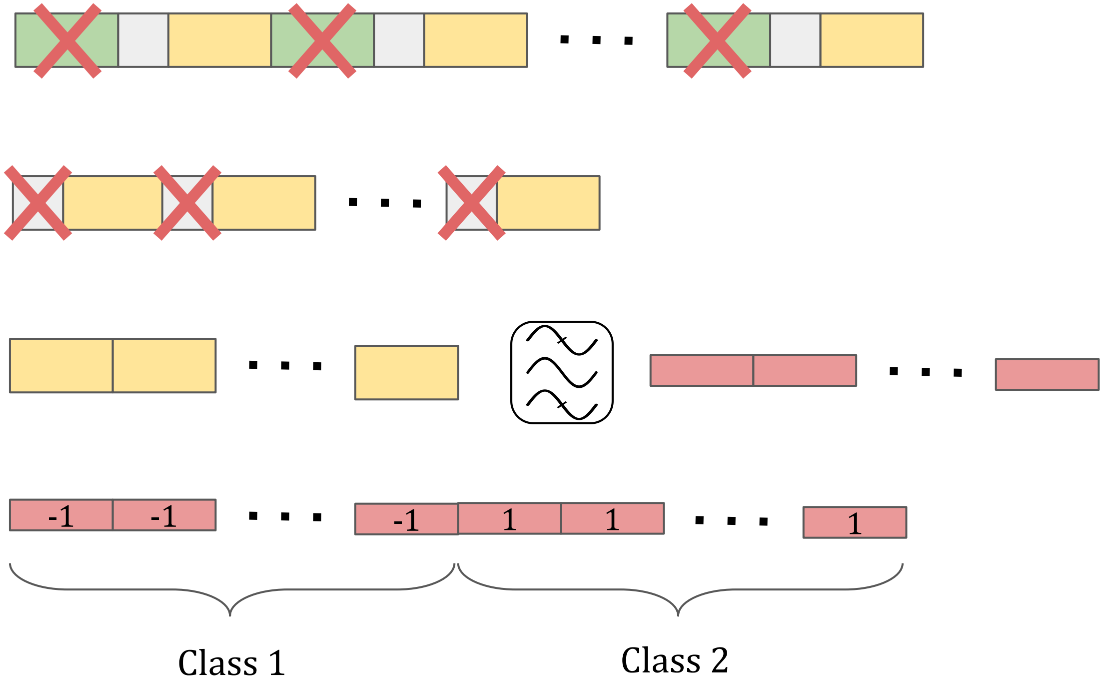
- Pre-processing

Remove neutral intervals

Remove intervals of visual stimulation

Apply band-pass filter

Merge two sets of user data



Methodology (Cont.)

- Feature extraction
    - Common Spatial Pattern (CSP) was used as a feature extraction algorithm.
    - Eigenvalue method was used to compute the CSP.

MATLAB Implementation

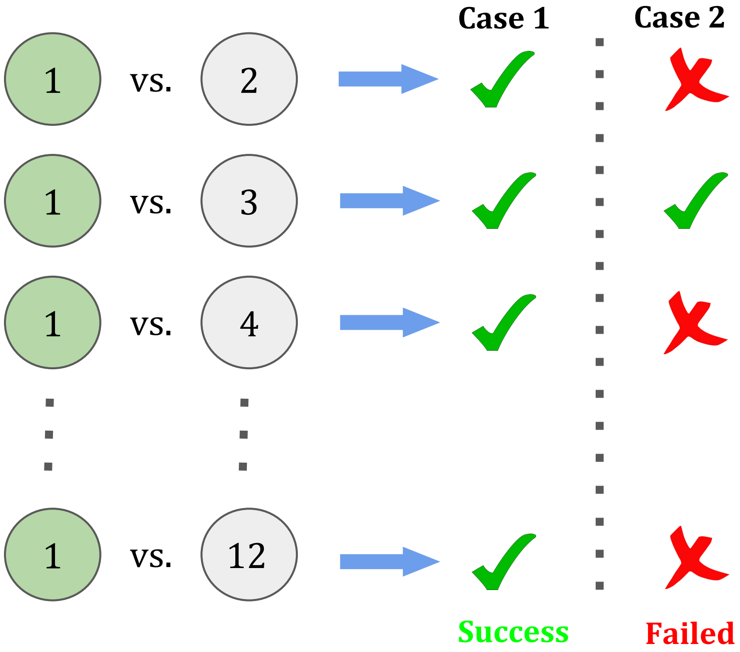
```
[V,D] = eig(cov(class2),cov(class1)+cov(class2));
```

Methodology (Cont.)

- Classification
  - Linear Discriminant Analysis (LDA) was used to classify pairs of users.
  - 10 trials for each user were used to train the model.
  - 5 trials for each user were used to test the model.
- Weights and bias values were calculated and recorded for each user pair.

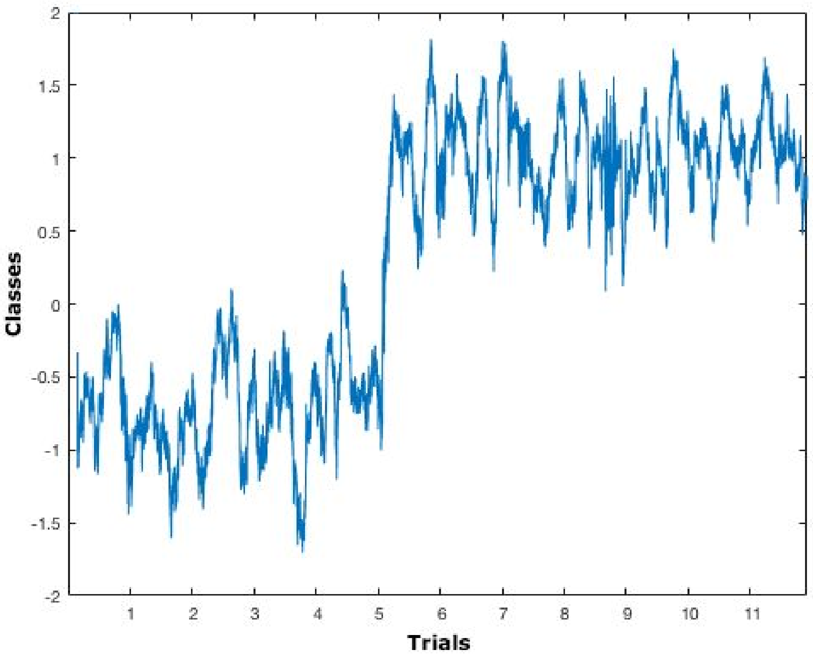
Methodology (Cont.)

- Authentication
    - A particular user is checked against other users.



Result and Discussion

- Classification result for two subjects (subjects 5 & 6)



Result and Discussion (Cont.)

- Number of possible pairs: $\frac{12 \times 11}{2} = 66$
- Tests were performed for three different frequency bands

Identifier	Frequency band	Correctly classifying pairs	Accuracy
Alpha, α	8 – 13 Hz	54	81.81%
Beta, β	13 – 30 Hz	60	90.91%
Alpha + Beta, $\alpha + \beta$	8 – 30 Hz	64	96.97%

Conclusion

- Combination of Alpha and Beta bands (8 – 30 Hz) showed better accuracy.
- Overall system accuracy is 66.67% because four subjects (2, 4, 9, and 10) failed to authenticate correctly.
- Accuracy can be improved by using additional features as well as changing machine learning algorithm.

References

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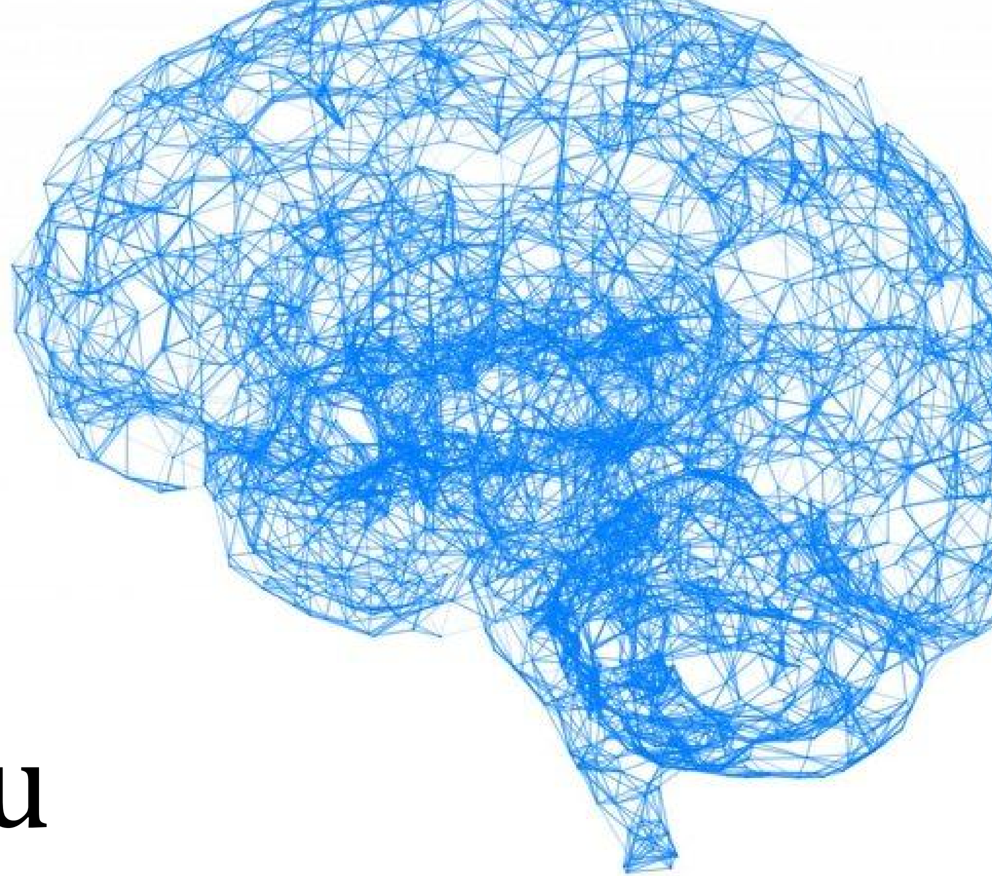

Thank you